

ΟΜΑΔΑ ΑΣΚΗΣΕΩΝ 10

Άσκηση 1

Καθίστα A: $K^A = \begin{bmatrix} 1 & 0 & c_x \\ 0 & 1 & c_y \\ 0 & 0 & 1 \end{bmatrix}$ $R^A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$t_A = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Ισχύει $P_A = K_A R_A [I | -t_A]$ $P_A = I$ Άρα $t_A = 0$

$P_A = K [I \ 0]$

$P_A = \begin{bmatrix} 1 & 0 & c_x & 0 \\ 0 & 1 & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

Ομοίως Για την Καθίστα B: $P_B = \begin{bmatrix} 1 & 0 & c_x & 0 \\ 0 & 1 & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$

ήτοι $P_B = I$ και $t_B = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Είμαστε σε ορθογώνιο σύστημα Άρα $X = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$

$X_A = P_A X \Rightarrow X_A = \begin{bmatrix} 1 & 0 & c_x & 0 \\ 0 & 1 & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} f \cdot X + c_x \cdot Z \\ f \cdot Y + c_y \cdot Z \\ Z \end{bmatrix} \begin{matrix} l_1 \\ l_2 \\ l_3 \end{matrix} \quad x = \frac{l_2}{l_3} \quad y = \frac{l_1}{l_3}$$

Determine l_i

$$x = \frac{f \cdot X + c_x \cdot Z}{Z} \quad y = \frac{f \cdot Y + c_y \cdot Z}{Z}$$

$$\textcircled{1} \Rightarrow \overset{x \cdot Z}{x \cdot Z} = f \cdot X + c_x \cdot Z \quad \textcircled{2} \Rightarrow \overset{y \cdot Z}{y \cdot Z} = f \cdot Y + c_y \cdot Z$$

$$\Rightarrow x \cdot Z - f \cdot X - c_x \cdot Z = 0 \quad y \cdot Z - f \cdot Y - c_y \cdot Z = 0$$

$$\text{Ans: } x \cdot Z - f \cdot X - c_x \cdot Z = 0 \quad y \cdot Z - f \cdot Y - c_y \cdot Z = 0$$

fin row B: $XA = P_0X \Rightarrow \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} f & 0 & c_x & f_b \\ 0 & f & c_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$

$$\Rightarrow \begin{bmatrix} f \cdot X + c_x \cdot Z + f_b \\ f \cdot Y + c_y \cdot Z \\ Z \end{bmatrix} \begin{matrix} l_1 \\ l_2 \\ l_3 \end{matrix} \quad x = \frac{l_2}{l_3} \quad y = \frac{l_1}{l_3}$$

$$\Rightarrow x = \frac{f \cdot X + c_x \cdot Z + f_b}{Z} \quad y = \frac{f \cdot Y + c_y \cdot Z}{Z}$$

$$\textcircled{1} \Rightarrow x \cdot Z - f \cdot X - c_x \cdot Z - f_b = 0 \quad y \cdot Z - f \cdot Y - c_y \cdot Z = 0$$

⑤

$$\textcircled{1} \Rightarrow \cancel{1}x - (y + (x)z) - (x - 0y - (x + (x)z) = 0$$

$$\textcircled{2} \Rightarrow -0x - 1y - (-y + (y)z) = 0$$

$$\textcircled{3} \Rightarrow -1x - 0y - (x - x) = +1b$$

$$\textcircled{4} \Rightarrow \textcircled{1} - \textcircled{2} - \textcircled{3} - \textcircled{4} \Rightarrow -0x - 1y - (y - y) = 0$$

$$\begin{bmatrix} -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x - x_0 \\ y - y_0 \\ x - x_0 \\ y - y_0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1b \\ 0 \end{bmatrix}$$

$$\textcircled{3} - \textcircled{1} \Rightarrow \sqrt{2} - (x - (x)z) + 1b - xz + (x + (x)z) = 0$$

$$\textcircled{1} \Rightarrow x = \frac{1x}{2} + (x) \Rightarrow x_0(x) = \frac{1x}{2} \textcircled{I}$$

$$\textcircled{1} \Rightarrow x_0(x) = \frac{1x}{2} + \frac{1b}{2} \textcircled{II}$$

$$\textcircled{II} - \textcircled{I} \Rightarrow (x_0(x) - (x_0(x))) = \frac{1b}{2} \Rightarrow x_0 - x_0 = \textcircled{0} \frac{1b}{2}$$

$$\Rightarrow \boxed{2 = \frac{1b}{x_0 - x_0}} \quad \textcircled{2} \Rightarrow x_0(x) = \frac{1x}{2} \Rightarrow$$

$$\Rightarrow \cancel{X_A - (x)} = \frac{X_B - X_A}{f + f_b} \Rightarrow$$

$$(\cancel{X - f_b})(X_A - (x)) = \cancel{X_B - X_A}$$

$$\cancel{f_x} = \frac{X_B - X_A}{f_b(X_A - (x))}$$

$$\Rightarrow (X_A - (x)) \cdot z = f_x \Rightarrow$$

$$x = \frac{z(X_A - (x))}{f} \Rightarrow$$

$$x = \frac{f_b}{X_B - X_A} (X_A - (x)) \Rightarrow$$

$$\Rightarrow \boxed{x = \frac{b(X_A - (x))}{X_B - X_A}}$$

$$(2) \Rightarrow \boxed{y = \frac{b(X_A - (y))}{X_B - X_A}}$$

Agrupación 2

Sumando:

→ A

Grupo 1: $X_1 = 862$ $Y_1 = 438$
 Grupo 2: $X_2 = 684$ $Y_2 = 485$
 Grupo 3: $X_3 = 1187$ $Y_3 = 388$
 Grupo 4: $X_4 = 1585$ $Y_4 = 568$

Mediana: $X = 266$ $Y = 155$
 Moda: $X = 336$ $Y = 868$

Supedit:

Aplicar Grupo 1: $X = 1614$ $Y = 521$
 Dejar Grupo 2: $X = 1453$ $Y = 318$

Agrupación B

Sumando:

Grupo 1: $X_1 = 614$ $X_2 = 792$
 Grupo 2: $X_2 = 723$ $Y_2 = 462$
 Grupo 3: $X_3 = 952$ $X_4 = 358$
 Grupo 4: $X_4 = 1296$ $X_5 = 540$

Mediana: $X = 54$ $Y = 230$
 Moda: $X = 202$ $Y = 870$

Supedit:

Aplicar Grupo 1: $X = 1470$
 $Y = 567$

Dejar: $X = 7353$ ~~$Y = 312$~~
 $Y = 312$

Latia 1

$$d = x_A - x_B = 962 - 634 = 328$$

↑ And d'amen 1 e z'itos

$$z = \frac{f \cdot B}{d} = \frac{175823 \cdot 111,53}{328} = 59785 \text{ mm}$$

$$x = \frac{(x_A - x) \cdot z}{f} = \frac{(962 - 963,34) \cdot 59785}{175823} =$$

$$= \frac{8,66 \cdot 59785}{175823} = 2,94 \text{ mm}$$

$$y = \frac{(y_A - y) \cdot z}{f} = \frac{(738 - 552,28) \cdot 59785}{175823} =$$

$$= \frac{185,71 \cdot 59785}{175823} = 63,15 \text{ mm}$$

$$\text{Apa } P_i = (2,94, 63,15, 59785)$$

Latia 2

$$d = x_A - x_B = 684 - 428 = 256$$

$$z = \frac{175823 \cdot 111,53}{256} = 76600 \text{ mm}$$

$$X = \frac{(684 - 953.34) \cdot 766}{1758.23} = -117.34 \text{ mm}$$

$$Y = \frac{(485 - 552.29) \cdot 766}{1758.23} = -29.32 \text{ mm}$$

Apda $P_2 = (-117.34, -29.32, 76600)$

Tump 3 :

$$d = 1187 - 952 = 235$$

$$Z = 834.45 \text{ mm}$$

$$X = 110.89 \text{ mm}$$

$$Y = -77.97 \text{ mm}$$

Apda $P_3 = (110.89, -77.97, 83445)$

Tump 4 :

$$d = 289$$

$$Z = 678.53 \text{ mm}$$

$$X = 243.77 \text{ mm}$$

$$Y = 606 \text{ mm}$$

Apda $P_4 = (243.77, 606, 678.53)$

~~Exercício~~
Xupijate co 4nãpo oi São João

$$\epsilon = \epsilon_{123} + \epsilon_{134}$$

$$\epsilon_{123} = \frac{1}{2} \|(P_2 - P_1) \cdot (P_3 - P_1)\|$$

Adicional $P_2 - P_1$: $\vec{a} = (-117,34 - 2,94, -79,32 - 69,15,$
 $766,00 - 537,85) \Rightarrow$
 $\vec{a} = (-120,28, -92,47, 168,15)$

Adicional $P_3 - P_1$: $\vec{b} = (11089 - 2,94, -77,97 - 69,15,$
 $831,45 - 537,85) \Rightarrow$
 $\vec{b} = (107,95, -141,12, 236,60)$

$$\vec{a} \cdot \vec{b} = \begin{vmatrix} i & j & k \\ -120,28 & -92,47 & 168,15 \\ 107,95 & -141,12 & 236,60 \end{vmatrix} \quad \#$$

Zerando i:

$$\begin{aligned} & (-92,47)(236,60) - (168,15)(-141,12) = -21878,40 + \\ & 23730,83 = \\ & -1852,43 \end{aligned}$$

Zerando j: $- [(-120,28)(236,60) - (168,15)(107,95)] =$
 $= - [-28481,28 - 18152,75] = 46571,03$

$$\text{Zuvorka } k: (-120.28)(-141.12) - (-92.47)(107.95) = \\ = 16972.47 + 9987.19 = 26959.66$$

$$\vec{a} \cdot \vec{b} = (1852.43, 46571.04, 26959.66)$$

$$\|\vec{a} \cdot \vec{b}\| = \sqrt{1852.43^2 + 46571.04^2 + 26959.66^2} = \\ = \sqrt{2893114497} = 53843.4$$

$$\epsilon_{123} = \frac{1}{2} \cdot 53843.4 = 26921.7 \text{ mm}^2$$

Διάφορα: $P_3 - P_1$:

$$\vec{c} = (107.95, -141.12, 236.60)$$

$$\text{Διάφορα } P_4 - P_1: \vec{d} = (243.77 - 294, 6.06 - 6315, \\ 678.53 - 597.85) \Rightarrow$$

$$\vec{d} = (240.83, -57.09, 80.68)$$

$$\vec{c} \cdot \vec{d} = \begin{vmatrix} i & j & k \\ 107.95 & -141.12 & 236.60 \\ 240.83 & -57.09 & 80.68 \end{vmatrix}$$

$$i = (-141.12)(80.68) - (236.60)(-57.09) = 2118.33$$

$$j = -[107.95(80.68) - 236.60(240.83)] = 48190.97$$

$$k = 107.95(-57.09) - (-141.12)(240.83) = 27821.83$$

$$\vec{z} \cdot \vec{d} = (2118.33, 48190.93, 27821.83)$$

$$\|\vec{z} \cdot \vec{d}\| = \sqrt{2118.33^2 + 48190.93^2 + 27821.83^2} = 55684.4$$

$$E_{134} = \frac{1}{2} \cdot 55684.4 = 27842.2 \text{ mm}^2$$

$$\text{Apa Ecuatiei pas} = E_{123} + E_{134} = 26921.7 + 27842.2 = 54763.9 \text{ mm}^2$$

Vipos kopētās:

$$P_{\text{papriekš}} = (-361.60, -209.01, 1361.77)$$

$$P_{\text{pāri}} = (-337.83, 191.36, 1056.74)$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} = 425.1 \text{ mm} \approx 42.5 \text{ cm}$$

Itāzies supriapaļ

$$P_{\text{papriekš}} = (311.69, 144.9, 1361.77)$$

$$P_{\text{pāri}} = (537.04, -246.51, 1056.96)$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} = 553.85 \text{ mm} \approx 55.4 \text{ cm}$$

③

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Asılman 3

$$P_A = \begin{bmatrix} -1 & 1 & -1 & 1 \\ -1 & -2 & -2 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \quad P_B = \begin{bmatrix} 2 & 0 & -1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\text{logun } x_A = P_A X \quad \text{ve } x_B = P_B X$$

$$\begin{pmatrix} x_A \\ y_A \\ 1 \end{pmatrix} = \begin{bmatrix} -1 & 1 & -1 & 1 \\ -1 & -2 & -2 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix} \Rightarrow$$

$$x_A = \frac{-1X + Y - Z + 1}{-2 + 1} \Rightarrow -2 + 1 = -X + Y - Z + 1 \Rightarrow \boxed{X - Y = 0} \quad (1)$$

$$y_A = \frac{-X - 2Y - 2Z + 2}{-2 + 1} \Rightarrow 2 = \frac{-X - 2Y - 2Z + 2}{-2 + 1} \Rightarrow -2 + 2 = -X - 2Y - 2Z + 2 \Rightarrow \boxed{X + 2Y = 0} \quad (2)$$

$$\text{Kırdığımızı öze } X = Y = 0$$

Ayrıca y_B kısmı B:

$$x_B = \frac{2X + 0Y - Z + 0}{0X + 0Y + Z + 1} \quad (3) \quad y_B = \frac{0X + 1Y + Z + 0}{0X + 0Y + Z + 1} \quad (4)$$

(2)

$$\textcircled{3} \Rightarrow -0.62 - 0.6 = 2X - 2 + 1 \Rightarrow \boxed{2X - 0.42 = -1.6}$$

$$\textcircled{4} \Rightarrow 0.82 + 0.8 = Y + 2 \Rightarrow Y + 2 - 0.82 = 0.8$$

$$\Rightarrow \boxed{Y + 0.22 = 0.8}$$

Exempl 4 System:

$$\begin{cases} 1X - 1Y + 0Z = 0 \\ 1X + 2Y + 0Z = 0 \\ 2X + 0Y - 0.42 = -1.6 \\ 0X + 1Y + 0.22 = 0.8 \end{cases} \quad \begin{array}{l} \text{Koeffizienten in Matrix} \\ Ax = b \end{array}$$

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 0 \\ 2 & 0 & -0.42 \\ 0 & 1 & 0.22 \end{bmatrix} \quad X = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix} \quad b = \begin{bmatrix} 0 \\ 0 \\ -1.6 \\ 0.8 \end{bmatrix}$$

$$\text{Lösung } X = (A^T A)^{-1} A^T b$$

$$A^T = \begin{bmatrix} 1 & 1 & 2 & 0 \\ -1 & 2 & 0 & 1 \\ 0 & 0 & -0.42 & 0.22 \end{bmatrix}$$

③

Bei Vorgehensweise zur $A^T A$, $A^T b$

$\det(A^T A)^{-1}$

$$A^T A = \begin{bmatrix} 1 & 1 & 2 & 0 \\ -1 & 2 & 0 & 1 \\ 0 & 0 & -0.4 & 0.2 \end{bmatrix} \begin{bmatrix} 1 & -1 & 0 \\ 1 & 2 & 0 \\ 2 & 0 & -0.4 \\ 0 & 1 & 0.2 \end{bmatrix}$$

3×4 4×3

$$\Rightarrow A^T A = \begin{bmatrix} 6 & 7 & -0.8 \\ 7 & 6 & 0.2 \\ -0.8 & 0.2 & 0.2 \end{bmatrix} \rightarrow 3 \times 3$$

$$A^T b = \begin{bmatrix} 1 & 1 & 2 & 0 \\ -1 & 2 & 0 & 1 \\ 0 & 0 & -0.4 & 0.2 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -1.6 \\ 0.8 \end{bmatrix}$$

3×4 4×1

$$\Rightarrow \begin{bmatrix} -3.2 \\ 0.8 \\ 0.8 \end{bmatrix}$$

3×1

~~$(A^T A)^{-1}$~~

~~$\det(A^T A) = 6 \cdot (6 \cdot 0.2 - 0.2 \cdot 0.2)$~~

$$\det(A^T A) = 6(6 \cdot 0.2 - 0.2 \cdot 0.2) - 7(0.2 + 0.16) - 0.8(0.2 \cdot 0.2)$$

$$= 6 \cdot 1.16 - 0.36 - 0.8 \cdot 5 = \boxed{-11.4}$$

(4)

$$+ (6 \cdot 0.2 - 0.2 \cdot 0.2) = 1.2 - 0.04 = 1.16$$

$$- (1 \cdot 0.2 - 0.2 \cdot (-0.8)) = -0.36$$

$$+ (1 \cdot 0.2 - 6 \cdot (-0.8)) = 5$$

$$C_{21} = -0.36$$

$$+ (6 \cdot 0.2 - 0.8 \cdot (-0.8)) = 1.2 - 0.64 = 0.56$$

$$- (6 \cdot 0.2 - 0.8 \cdot (-0.8)) = -0.56$$

$$- (6 \cdot 0.2 - 1 \cdot (-0.8)) = - (1.2 + 0.8) = -2$$

$$C_{22} = 0.56$$

$$C_{23} = -2$$

$$C_{31} = + (6 \cdot 6 - 1 \cdot 1) = 35$$

$$(A^T A)^{-1} = \frac{1}{-14.9} \begin{bmatrix} 1.16 & -0.36 & 5 \\ 0.36 & 0.56 & -2 \\ 5 & -2 & 35 \end{bmatrix}$$

$$(A^T A)^{-1} = \begin{bmatrix} -0.0866 & 0.0267 & -0.3311 \\ 0.0267 & -0.0478 & 0.1393 \\ -0.3311 & 0.1393 & -2.6111 \end{bmatrix}$$

$$\hat{x} = \begin{bmatrix} -0.0866 & 0.0267 & -0.3311 \\ 0.0267 & -0.0478 & 0.1393 \\ -0.3311 & 0.1393 & -2.6111 \end{bmatrix} \cdot \begin{bmatrix} -3.2 \\ 0.8 \\ 0.8 \end{bmatrix}$$

$(A^T A)^{-1}$ 3×3 3×1 $A^T b$